

## GCSE algebra - Set 1

TMUA practice questions by topic. Specification M4.1-M4.8, M4.18, M4.19. 12 questions.

Each question has one correct answer. No calculator. Answers and worked solutions are in the separate solutions PDF.

- 1 Make  $x$  the subject of  $y = \frac{2x+3}{x-1}$ .
- A**  $x = \frac{3-y}{y-2}$   
**B**  $x = \frac{y-2}{y \pm 3}$   
**C**  $x = \frac{y \pm 3}{y \mp 3}$   
**D**  $x = \frac{y \mp 3}{y \mp 3}$   
**E**  $x = \frac{y \mp 3}{2y+3}$   
**F**  $x = \frac{2y+3}{y-1}$
- 2 Find the  $n$ th term of the quadratic sequence 3, 8, 15, 24, 35, ...
- A**  $n^2 + 2n$   
**B**  $(n+1)^2 + 1$   
**C**  $n^2 + 3n - 1$   
**D**  $n^2 + 2$   
**E**  $n^2 + n + 1$   
**F**  $2n^2 + 1$
- 3 Which of the following is an identity (true for all values of  $x$ )?
- A**  $(x+3)^2 - (x-3)^2 = 12x$   
**B**  $x^2 + 6x + 9 = 0$   
**C**  $(x+3)^2 = x^2 + 9$   
**D**  $(x-3)^2 = x^2 - 6x - 9$   
**E**  $(x+3)(x-3) = x^2 + 9$

- 4 A car accelerates uniformly from rest to 20 m/s in 8 seconds, travels at 20 m/s for 12 seconds, then decelerates uniformly to rest in 10 seconds. Find the total distance travelled.

A 600 m  
 B 400 m  
 C 420 m  
 D 300 m  
 E 380 m  
 F 480 m

- 5 Given that  $v^2 = u^2 + 2as$ , find  $s$  when  $v = 13$ ,  $u = 5$  and  $a = 4$ .

A 18  
 B 21  
 C 36  
 D 12  
 E 9  
 F 24

- 6 Simplify  $\frac{3}{x+2} - \frac{2}{x-1}$ .

A  $\frac{3x-4}{(x+2)(x-1)}$   
 B  $\frac{x-4}{(x+2)(x-1)}$   
 C  $\frac{x+3}{x-7}$   
 D  $\frac{x+2}{(x+2)(x-1)}$   
 E  $\frac{x+2}{(x+2)(x-1)}$   
 F  $\frac{x+2}{(x+2)(x-1)}$

- 7 Given that  $T = 2\pi\sqrt{\frac{l}{g}}$ , express  $l$  in terms of  $T$  and  $g$ .

A  $l = \frac{gT}{2\pi^2}$   
 B  $l = \frac{4\pi^2 g}{T^2}$   
 C  $l = \frac{4\pi^2 g}{T^2}$   
 D  $l = \frac{4\pi^2 g}{T^2}$   
 E  $l = \frac{4\pi^2 g}{T^2}$   
 F  $l = 4\pi^2 g T^2$

- 8** A ball is thrown vertically upwards and its height after  $t$  seconds is  $h = 20t - 5t^2$  metres. For how long is the ball at least 15 m above the ground?
- A** 4 s  
**B** 1 s  
**C** 0.5 s  
**D** 1.5 s  
**E** 2 s  
**F** 3 s
- 9** Expand and simplify  $(x + 2)(x - 1)(x + 3)$ .
- A**  $x^3 + 4x^2 + 7x - 6$   
**B**  $x^3 + 2x^2 - 5x - 6$   
**C**  $x^3 - 4x^2 + x + 6$   
**D**  $x^3 + 4x^2 - x - 6$   
**E**  $x^3 + 6x^2 + 11x + 6$   
**F**  $x^3 + 4x^2 + x - 6$
- 10** A sequence has first term 2 and each subsequent term is obtained by squaring the previous term and subtracting 2. What is the 50th term?
- A**  $2^{50}$   
**B**  $-2$   
**C** 14  
**D** 4  
**E** 0  
**F** 2

**11** Factorise  $6x^2 - 7x - 20$ .

**A**  $(3x - 4)(2x + 5)$

**B**  $(6x - 5)(x + 4)$

**C**  $(3x + 4)(2x - 5)$

**D**  $(3x - 5)(2x + 4)$

**E**  $(6x + 5)(x - 4)$

**F**  $(3x + 5)(2x - 4)$

**12** The sum of two numbers is 12 and the difference of their squares is 48. Find the larger number.

**A** 6

**B** 10

**C** 8

**D** 7

**E** 4

**F** 9